

Enterprise Google Cloud Data Platform Blueprint

Volume 03: Enterprise Data Platform

Data sources, ingestion, storage, governance and data products

Document attribute	Value
Version	0.1 reference implementation
Date	2026-07-10
Intended audience	Executive, architecture, security, platform engineering, data, AI, SRE and FinOps stakeholders
Scope	Fortune 100 reference blueprint for petabyte-scale Google Cloud data platform
Status	Generated implementation starter set; verify service-specific settings against current Google Cloud documentation

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Inherited Fortune 100 Assumptions

Area	Assumption
Scale	10-50 PB historical data; 50-250 TB/day new data; billions of events/day
Organization	Global enterprise with central platform team and federated domain teams
Regulatory	GDPR, HIPAA, PCI DSS, SOC 2, ISO 27001, NIST and regional residency constraints may apply
Availability	Tier 0 RTO <1 hour and RPO <5 minutes; Tier 1 RTO <4 hours and RPO <15 minutes
Cloud posture	Hybrid and multi-cloud integration with Google Cloud as the enterprise data platform foundation
Security	Zero Trust, least privilege, private connectivity, service perimeters, CMEK for restricted data

1. Platform Objectives

The enterprise data platform unifies batch, streaming, CDC, API and SaaS data into governed data layers and certified data products.

2. Data Layers

Landing, quarantine, raw/bronze, silver, gold/curated, semantic, sandbox and archive layers provide clear quality, lifecycle and ownership boundaries.

Layer	Purpose	Primary Services	Controls
Landing	Transient source handoff	Cloud Storage	Retention, checksum, malware scan
Raw/Bronze	Immutable source history	Cloud Storage/BigQuery	Restricted access, lineage
Silver	Standardized and conformed	BigQuery	DQ checks, schema contracts
Gold/Curated	Certified business data products	BigQuery/BigLake	Owner, SLO, policy tags
Archive	Long-term retention/legal hold	Cloud Storage	Lifecycle and retention policies

Figure: Enterprise Data Platform - Data Layers

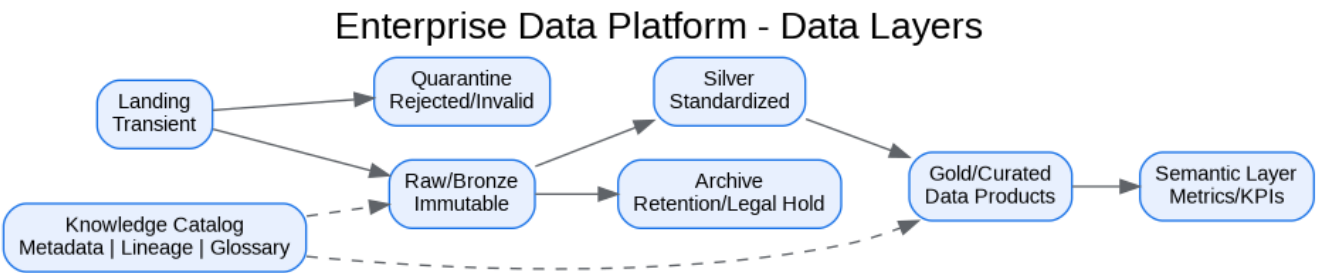
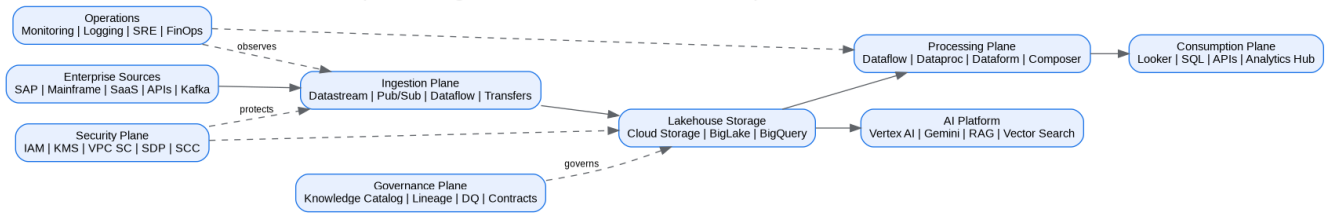


Figure: Enterprise Google Cloud Data Platform - Conceptual Reference Architecture

Enterprise Google Cloud Data Platform - Conceptual Reference Architecture



3. Ingestion Architecture

Batch ingestion uses managed transfers and Dataflow; CDC uses Datastream; event ingestion uses Pub/Sub and Dataflow; API/SaaS ingestion uses connector patterns and Cloud Run when custom integration is needed.

4. Metadata and Governance

Knowledge Catalog provides enterprise metadata, business glossary, lineage, quality status, certification, data-product registration and discovery.

5. Service Assessment

Services are selected only where they support business capabilities, operational controls or compliance requirements.

Service	Purpose	Benefits	Limitations / Pitfalls	Cost Controls
BigQuery	Enterprise analytics warehouse and curated serving layer	Serverless SQL, separation of storage/compute, strong ecosystem	Poor partitioning and unrestricted ad-hoc queries can create high costs	Reservations, labels, max bytes billed, partitioning, clustering
Cloud Storage	Raw, landing, archive and open-format lakehouse storage	Durable object storage, lifecycle policies, broad integration	Bucket sprawl and direct consumer access can weaken governance	Lifecycle policies, storage classes, retention controls
BigLake	Governed access to Cloud Storage data through BigQuery interfaces	Fine-grained access without direct bucket reads	Requires metadata and permission design	Table lifecycle, open-format governance
Dataflow	Batch and streaming pipelines using Apache Beam	Unified model, autoscaling, managed execution	Complex stateful pipelines require strong testing	Autoscaling, worker sizing, streaming engine
Pub/Sub	Event ingestion and decoupling	Reliable async messaging, retry and dead-letter support	Schema/version drift can impact consumers	Retention tuning and payload design
Knowledge Catalog	Metadata, glossary, lineage, governance and data discovery	Enterprise context layer for data and AI	Incomplete stewardship reduces trust	Metadata ownership and lifecycle controls

Architecture Decision Records

ADR	Decision	Problem	Alternatives	Decision	Tradeoffs/Risks	Mitigation
ADR-001	Google Cloud as enterprise data platform	Fragmented analytics estate	Continue legacy; multi-cloud split; Google Cloud data platform	Adopt Google Cloud data platform with hybrid integration	Requires operating-model change	Phased landing zone, domain pilots, and governance automation
ADR-002	BigQuery analytical core	Need PB-scale SQL analytics	Legacy EDW; Spark-only;	Use BigQuery for	Cost governance required	Reservations, labels, query

			third-party warehouse	curated/serving layers		controls, FinOps dashboards
ADR-003	Lakehouse with Cloud Storage and BigLake	Need open data and governed access	BigQuery-only; unmanaged object lake	Use Cloud Storage + BigLake + BigQuery	Open table governance maturity required	Catalog, policies, certified data-product lifecycle
ADR-004	Zero Trust and VPC SC	Exfiltration risk and regulated data	Network-only perimeter; IAM-only controls	Use identity, network, data and service perimeters	Service exceptions require testing	Perimeter dry runs and explicit ingress/egress policies

Risks and Mitigations

Risk	Impact	Mitigation
Underestimated legacy complexity	High	Complete source, pipeline, report and dependency assessment before migration waves
Data ownership gaps	High	Create domain owner RACI and data-product certification gate
Cost overrun	High	Enforce labels, budgets, reservations, query guardrails and monthly FinOps reviews
Security control drift	Critical	Deploy IaC, policy-as-code, automated IAM review and security posture monitoring
AI data leakage	Critical	Use certified datasets, service perimeters, DLP, RAG guardrails and model governance

Production Readiness Checklist

- All projects, networks, KMS keys and service accounts deployed through Terraform.
- All data products registered with owner, steward, classification, SLO, contract and support model.
- All restricted paths validated against VPC Service Controls and private connectivity requirements.
- All pipelines implement retries, dead-letter handling, reconciliation and operational metrics.
- All dashboards, alerts, runbooks and escalation routes tested before production cutover.
- All cost allocation labels present and billing export dashboards available to product owners.

References

- Google Cloud Architecture Framework: <https://docs.cloud.google.com/architecture/framework>
- Google Cloud Enterprise Foundations Blueprint: <https://docs.cloud.google.com/architecture/blueprints/security-foundations>
- Google Cloud Terraform blueprints and modules: <https://docs.cloud.google.com/docs/terraform/blueprints/terraform-blueprints>
- Build data products in a data mesh: <https://docs.cloud.google.com/architecture/build-data-products-data-mesh>
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- Knowledge Catalog formerly Dataplex: <https://cloud.google.com/products/knowledge-catalog>
- Optimize costs with FinOps hub: <https://docs.cloud.google.com/billing/docs/how-to/finops-hub>
- VPC Service Controls supported products: <https://docs.cloud.google.com/vpc-service-controls/docs/supported-products>